



Assignment No. 01(Graded)
Semester: Spring 2026
CS304- Object Oriented Programming

Total Marks: 20

Due Date: 30th April , 2026

Uploading instructions:

- Your assignment should be in .doc or .docx files (Any other formats like scan images, PDF, zip, rar and bmp, etc. will not be accepted).
- Save your assignment with your ID (e.g. bc000000000.doc).
- No assignment will be accepted through email.

Rules for Marking:

It should be clear that your assignment will not get any credit if:

- The assignment is submitted after the due date.
- The submitted assignment does not open, or the file is corrupted.
- Your assignment is copied from the internet, handouts, or any other student. (Strict disciplinary action will be taken in this case).

Lectures Covered:

This assignment covers **Lectures # 1 to 7**

Note: Submit your solutions in .doc or .docx file.

Assignment No. 1

Objectives:

The objectives and learning outcomes of this assignment are to:

- Learn the main objects from a problem statement.
- Learn the necessary attributes and behaviors associated with each object.
- Learn the relationships among objects.
- Construct a comprehensive class diagram.

Learning Outcomes:

After completing this assignment, students will be able to:

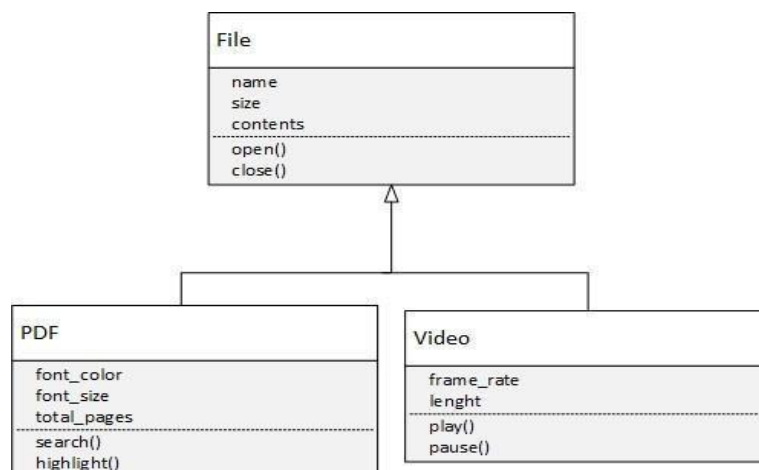
- Develop a UML class diagram.

Problem Statement:

A company is developing an **Online Food Delivery System** similar to Foodpanda. In this system, different types of users interact with the platform, all sharing some common details such as name, email, and login credentials, while also having their own specific functionalities, where this shared information is maintained in a unified manner and extended by specific user roles such as customers and delivery riders. Customers can create accounts, browse restaurants, select food items, and place orders. Each order is linked to one customer and one restaurant and includes multiple food items that are created specifically for that order and do not exist independently once the order is removed, and the payment information is also generated strictly as part of the order and cannot exist without it. Restaurants receive and process orders, updating their status accordingly, while a delivery rider is assigned to deliver each order and update delivery progress. The system also manages payments, which may be completed online or through cash on delivery, ensuring all order-related elements are tightly bound to the order and are created and destroyed along with it.

Develop a UML class diagram by identifying objects, their attributes (Characteristics) and Operations.

Sample output for the program is given below, your solution output should be like this.



Best of luck!

NOTE: Do not put any query on MDB about this assignment, if you have any query then email at cs304@vu.edu.pk. Furthermore, if any student is found cheating by any other student or from online forums then he/she will be awarded ZERO right away and strict disciplinary action will be taken against the student.

Deadline: Your assignment must be uploaded/submitted on or before 30th April 2026.